Derivative of the Variational Energy

Deriving the Monte Carlo Gradient Formula

Morten Hjorth-Jensen

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Setup and notation

Let $\psi_{\boldsymbol{\alpha}}(x)$ be a (real-valued) variational wavefunction depending on parameters $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_P)$ and x denote the many-body coordinates (continuous or discrete).

Variational energy

$$E(\boldsymbol{\alpha}) = \frac{\langle \psi_{\boldsymbol{\alpha}} | H | \psi_{\boldsymbol{\alpha}} \rangle}{\langle \psi_{\boldsymbol{\alpha}} | \psi_{\boldsymbol{\alpha}} \rangle} = \frac{\int dx\, \psi_{\boldsymbol{\alpha}}(x)\, [H\psi_{\boldsymbol{\alpha}}](x)}{\int dx\, |\psi_{\boldsymbol{\alpha}}(x)|^2}.$$

Define the normalization:

$$\mathcal{N}(\boldsymbol{\alpha}) := \int dx\, |\psi_{\boldsymbol{\alpha}}(x)|^2.$$

Local energy and sampling distribution

Introduce the **local energy**

$$E_L(x;\boldsymbol{\alpha}) := \frac{[H\psi_{\boldsymbol{\alpha}}](x)}{\psi_{\boldsymbol{\alpha}}(x)}.$$

Define the normalized probability density

$$p_{\boldsymbol{\alpha}}(x) := \frac{|\psi_{\boldsymbol{\alpha}}(x)|^2}{\mathcal{N}(\boldsymbol{\alpha})}, \qquad \int dx\, p_{\boldsymbol{\alpha}}(x) = 1.$$

Then the energy can be written as an expectation value:

$$E(\boldsymbol{\alpha}) = \int dx\, p_{\boldsymbol{\alpha}}(x)\, E_L(x;\boldsymbol{\alpha}) = \langle E_L \rangle\,.$$

Monte Carlo viewpoint

We estimate $E(\boldsymbol{\alpha})$ by sampling $x \sim p_{\boldsymbol{\alpha}}$ and averaging $E_L(x)$.

Differentiate $E(\boldsymbol{\alpha})$ in quotient form

Start from

$$E(\boldsymbol{\alpha}) = \frac{\int dx\, \psi_{\boldsymbol{\alpha}}(x)\, [H\psi_{\boldsymbol{\alpha}}](x)}{\int dx\, \psi_{\boldsymbol{\alpha}}(x)^2}$$

(assume $\psi_{\boldsymbol{\alpha}}$ real; complex case is similar with conjugation).
Let

$$A(\boldsymbol{\alpha}) := \int dx\, \psi_{\boldsymbol{\alpha}}\, H\psi_{\boldsymbol{\alpha}}, \qquad B(\boldsymbol{\alpha}) := \int dx\, \psi_{\boldsymbol{\alpha}}^2.$$

Then $E = A/B$ and

$$\partial_\alpha E = \frac{(\partial_\alpha A)B - A(\partial_\alpha B)}{B^2}.$$

Compute $\partial_\alpha A$ and $\partial_\alpha B$

Compute

$$\partial_\alpha B = \partial_\alpha \int dx\, \psi_{\boldsymbol{\alpha}}^2 = \int dx\, 2\psi_{\boldsymbol{\alpha}}\, (\partial_\alpha \psi_{\boldsymbol{\alpha}}).$$

For A:

$$\partial_\alpha A = \partial_\alpha \int dx\, \psi_{\boldsymbol{\alpha}}\, (H\psi_{\boldsymbol{\alpha}}) = \int dx \left[(\partial_\alpha \psi_{\boldsymbol{\alpha}})(H\psi_{\boldsymbol{\alpha}}) + \psi_{\boldsymbol{\alpha}}\, H(\partial_\alpha \psi_{\boldsymbol{\alpha}}) \right].$$

Use Hermiticity of H

Assuming suitable boundary conditions, H is Hermitian:

$$\int dx\, \psi_{\boldsymbol{\alpha}}\, H(\partial_\alpha \psi_{\boldsymbol{\alpha}}) = \int dx\, (\partial_\alpha \psi_{\boldsymbol{\alpha}})\, H\psi_{\boldsymbol{\alpha}}.$$

Thus

$$\partial_\alpha A = 2 \int dx\, (\partial_\alpha \psi_{\boldsymbol{\alpha}})\, H\psi_{\boldsymbol{\alpha}}.$$

Insert into quotient rule

Insert $\partial_\alpha A$ and $\partial_\alpha B$ into

$$\partial_\alpha E = \frac{(\partial_\alpha A)B - A(\partial_\alpha B)}{B^2}.$$

Using $A = EB$:

$$\partial_\alpha E = \frac{2\left(\int dx\,(\partial_\alpha \psi_{\boldsymbol{\alpha}})\,H\psi_{\boldsymbol{\alpha}}\right)B - (EB)\left(\int dx\,2\psi_{\boldsymbol{\alpha}}\,\partial_\alpha\psi_{\boldsymbol{\alpha}}\right)}{B^2}.$$

Cancel B:

$$\partial_\alpha E = 2\left[\frac{\int dx\,(\partial_\alpha\psi_{\boldsymbol{\alpha}})\,H\psi_{\boldsymbol{\alpha}}}{\int dx\,\psi_{\boldsymbol{\alpha}}^2} - E\,\frac{\int dx\,\psi_{\boldsymbol{\alpha}}\,\partial_\alpha\psi_{\boldsymbol{\alpha}}}{\int dx\,\psi_{\boldsymbol{\alpha}}^2}\right].$$

Rewrite using $p_\alpha(x)$ and local energy

Divide and multiply by $\psi_{\boldsymbol{\alpha}}$:

$$\frac{\int dx\,(\partial_\alpha \psi_{\boldsymbol{\alpha}})\,H\psi_{\boldsymbol{\alpha}}}{\int dx\,\psi_{\boldsymbol{\alpha}}^2} = \frac{\int dx\,\psi_{\boldsymbol{\alpha}}^2 \left(\frac{\partial_\alpha \psi_{\boldsymbol{\alpha}}}{\psi_{\boldsymbol{\alpha}}}\right)\left(\frac{H\psi_{\boldsymbol{\alpha}}}{\psi_{\boldsymbol{\alpha}}}\right)}{\int dx\,\psi_{\boldsymbol{\alpha}}^2}.$$

Recognize:

$$\frac{H\psi_{\boldsymbol{\alpha}}}{\psi_{\boldsymbol{\alpha}}} = E_L, \qquad \frac{\partial_\alpha \psi_{\boldsymbol{\alpha}}}{\psi_{\boldsymbol{\alpha}}} = \partial_\alpha \ln \psi_{\boldsymbol{\alpha}}.$$

Thus:

$$\partial_\alpha E = 2\left[\left\langle \left(\frac{\partial_\alpha \psi_{\boldsymbol{\alpha}}}{\psi_{\boldsymbol{\alpha}}}\right) E_L \right\rangle - E\left\langle \frac{\partial_\alpha \psi_{\boldsymbol{\alpha}}}{\psi_{\boldsymbol{\alpha}}} \right\rangle\right], \qquad E = \langle E_L \rangle\,.$$

Final covariance form

Since $E = \langle E_L \rangle$,

$$\partial_\alpha E = 2\left(\left\langle \left(\frac{\partial_\alpha \psi_{\boldsymbol{\alpha}}}{\psi_{\boldsymbol{\alpha}}}\right) E_L \right\rangle - \left\langle \frac{\partial_\alpha \psi_{\boldsymbol{\alpha}}}{\psi_{\boldsymbol{\alpha}}} \right\rangle \langle E_L \rangle\right).$$

Compact form (covariance)

Define the "log-derivative observable"

$$O_\alpha(x) := \frac{\partial_\alpha \psi_{\boldsymbol{\alpha}}(x)}{\psi_{\boldsymbol{\alpha}}(x)} = \partial_\alpha \ln \psi_{\boldsymbol{\alpha}}(x),$$

then

$$\boxed{\partial_\alpha E = 2\,\mathrm{Cov}(O_\alpha, E_L) = 2\left(\langle O_\alpha E_L \rangle - \langle O_\alpha \rangle \langle E_L \rangle\right).}$$

The three expectation values / integrals

Written explicitly in terms of $p_{\boldsymbol{\alpha}}(x)$, the gradient contains:

$$\langle O_\alpha E_L \rangle = \int dx\, p_{\boldsymbol{\alpha}}(x) \left(\frac{\partial_\alpha \psi_{\boldsymbol{\alpha}}(x)}{\psi_{\boldsymbol{\alpha}}(x)} \right) E_L(x),$$

$$\langle O_\alpha \rangle = \int dx\, p_{\boldsymbol{\alpha}}(x) \left(\frac{\partial_\alpha \psi_{\boldsymbol{\alpha}}(x)}{\psi_{\boldsymbol{\alpha}}(x)} \right),$$

$$\langle E_L \rangle = \int dx\, p_{\boldsymbol{\alpha}}(x)\, E_L(x).$$

Why this is useful

Each integral is an expectation value with respect to the *same* sampling distribution $p_{\boldsymbol{\alpha}}(x) \propto \psi_{\boldsymbol{\alpha}}(x)^2$, so all three are estimated from the *same* Monte Carlo run.

Practical Monte Carlo estimator

Given samples $\{x^{(m)}\}_{m=1}^{M} \sim p_{\boldsymbol{\alpha}}$, define

$$O_\alpha^{(m)} := \left.\frac{\partial_\alpha \psi_{\boldsymbol{\alpha}}}{\psi_{\boldsymbol{\alpha}}}\right|_{x^{(m)}}, \qquad E_L^{(m)} := E_L(x^{(m)}).$$

Then the unbiased (up to sampling error) estimator is

$$\widehat{\partial_\alpha E} = 2\left(\frac{1}{M}\sum_{m=1}^{M} O_\alpha^{(m)} E_L^{(m)} - \left(\frac{1}{M}\sum_{m=1}^{M} O_\alpha^{(m)}\right)\left(\frac{1}{M}\sum_{m=1}^{M} E_L^{(m)}\right)\right).$$

Interpretation

The gradient is proportional to the covariance between the log-derivative of the wavefunction and the local energy.

Remarks and extensions

- If $\psi_{\boldsymbol{\alpha}}$ is complex, replace $\psi_{\boldsymbol{\alpha}}^2$ by $|\psi_{\boldsymbol{\alpha}}|^2$ and

$$O_\alpha \equiv \partial_\alpha \ln \psi_{\boldsymbol{\alpha}} \quad \text{is generally complex.}$$

  The physically relevant gradient involves the *real part*:

$$\partial_\alpha E = 2\ \mathrm{Re}\left(\langle O_\alpha^* E_L\rangle - \langle O_\alpha^*\rangle \langle E_L\rangle\right).$$

- The same derivation applies to excited-state functionals and generalized cost functions.
- This identity underlies stochastic reconfiguration / natural gradient methods in VMC.

Summary

- Start from the Rayleigh quotient $E = \langle\psi|H|\psi\rangle/\langle\psi|\psi\rangle$.
- Differentiate using quotient rule and Hermiticity of H.
- Introduce $p(x) \propto |\psi(x)|^2$ and the local energy $E_L = (H\psi)/\psi$.
- Obtain the VMC gradient:

$$\partial_\alpha E = 2\left(\langle O_\alpha E_L\rangle - \langle O_\alpha\rangle\langle E_L\rangle\right).$$